

Speaker script

Extremality shift of rotating black holes at finite Gauss–Bonnet coupling

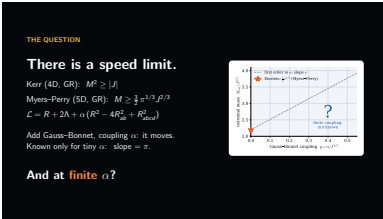
Five minutes, five slides plus a closing one: about 650 words at a calm pace. The times are targets; the elapsed time is in brackets. Paragraphs marked “Background” are for you, not part of the script. [click] marks a click within a slide: slide 3 has two steps and slide 4 three. Slides A1–A5 are backup for questions and are not part of the five minutes; their notes explain the method and the physics in more detail.



1. Title

0:15 [0:15]

This is joint work by Cecilia Bejarano, David Blanco, Guillem Pérez-Nadal and myself. Most of the code, the computations and the writing were done by two AI agents, Claude Code and Codex, under our direction.



2. Problem: the question

1:05 [1:20]

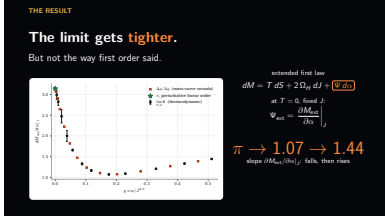
A rotating black hole cannot spin arbitrarily fast for its mass. For Kerr the bound is $J \leq M^2$. When it is saturated the temperature drops to zero, and the black hole is called *extremal*.

In five dimensions with two equal angular momenta the bound reads: M at least three halves $\pi^{1/3} J^{2/3}$. That is the orange star.

This bound is a property of Einstein’s equations. If we add the Gauss–Bonnet term, the natural first higher-curvature correction, with a coupling α — the bracket in the Lagrangian on the slide — the bound moves. How far? To first order in α the answer was known: the extremal mass grows with slope exactly π , the dashed line. Our question was what happens when α is *not* small, that is, the question mark.

Background: the action (for you, not part of the timed script; use a sentence of it if the audience needs it). In general relativity the dynamical field is the metric $g_{\mu\nu}$. The Einstein–Hilbert action, $\int \sqrt{-g} R$, with R the Ricci scalar (the simplest scalar built from the curvature), gives as Euler–Lagrange equations Einstein’s equations, $G_{\mu\nu} = 0$ in vacuum. They are second order in derivatives of the metric, like Newton’s or Maxwell’s equations. The Gauss–Bonnet term, $L_{\text{GB}} = R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$, is quadratic in the curvature, so it brings higher-derivative terms into the action. Terms like this are expected for two reasons. In effective field theory, general relativity is the first term of an expansion in curvature, and these are the first corrections; their size is set by the coupling α , which has units of length squared. They also

appear in the low-energy limit of string theory. The Gauss–Bonnet combination is special in two ways: its field equations stay second order, and in four dimensions it does not change the equations at all (it is a total derivative). That is one reason to work in five dimensions. The cosmological constant in the Lagrangian on the slide is zero throughout this work: the black holes are asymptotically flat.



3. Main result

1:20 [2:40]

The answer first: the limit gets tighter, but not the way first order said.

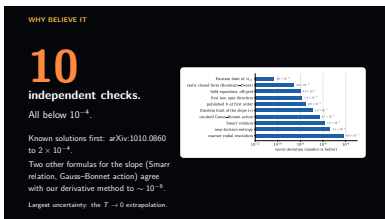
The key idea is thermodynamic, on the right of the slide. If we treat α as one more thermodynamic variable, the first law gets a new term, $\Psi d\alpha$. At zero temperature the $T dS$ term disappears, so at fixed spin Ψ is exactly the slope we want. We compute Ψ on every numerical solution, 356 of them at 13 couplings, and extrapolate to zero temperature.

The plot is that slope against the scaled coupling $y = \alpha/J^{2/3}$. At $y = 0$ we recover π , the green star. Then the slope falls fast, to about 1.07, and rises again to 1.44 at the largest coupling we reached: π to 1.07 to 1.44. The black dots are the thermodynamic values. The open squares are differences of the extremal mass between neighbouring couplings: a cross-check that does not use the entropy formula, though it shares the same solutions and extrapolation.

Three messages. The slope is always positive, so Gauss–Bonnet makes the spin bound tighter. It is far from constant, so the first-order formula fails quickly. And it is not monotonic.

[click] Now the extremal mass itself. It grows by 31 percent over the range, and the first-order line predicts a shift 2.3 times larger than the real one.

Background: the open squares. With $\mu = M_{\text{ext}}/J^{2/3}$, each square is the change in μ divided by the change in y between two neighbouring couplings, plotted at the midpoint: an interval average, not a pointwise slope, which is why the squares sit slightly off the dots near the minimum. They test the thermodynamic route (first law, Wald entropy, linear solve) because they use only masses read off at infinity. They are not independent of it: both use the same solutions and the same zero-temperature extrapolation, so an error in that extrapolation would move both alike. Worst interval: 1.6×10^{-4} , 0.2 percent of the change in μ over that interval (manuscript, “identity” equation).



4. Validation

1:10 [3:50]

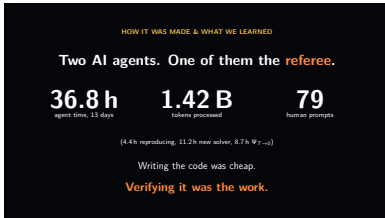
How do we know it is right? Ten independent checks, each shown by its worst deviation, and all of them below one part in ten thousand.

[click] Known solutions first. Before computing anything new, the code reproduced a published family of rotating Gauss–Bonnet black holes, from arXiv:1010.0860: lines ours, circles theirs, agreeing at the horizon to two parts in ten thousand. It also recovers the exact static solution and the value π at small coupling.

[click] Then independent routes. Ψ can also be obtained from the Smarr relation, using only the charges of a single solution, or as the integral of the Gauss–Bonnet density outside the black hole. Neither uses our derivative method, and both agree with it to about one part in a million.

The honest caveat, at the bottom of the slide. We never build a zero-temperature solution; we get very cold and extrapolate. That extrapolation is the largest uncertainty, and it is what the error bars in the result show.

Background: “all below 10^{-4} ” refers to the ten checks in the chart, the worst being 6.9×10^{-5} (coarser radial resolution). The comparison with arXiv:1010.0860 is not one of the ten: its 2×10^{-4} is limited by the resolution of their plot, not by our solver (see A1).



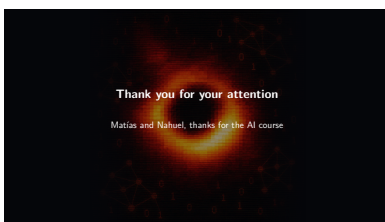
5. Experience: how it was made

1:05 [4:55]

Finally, how it was made. Two AI agents, and one of them was the referee. We chose the question and kept doubting; Claude Code wrote and ran the code; Codex acted as a referee and asked for checks. The referee rounds mattered: one check a referee asked for actually *failed* when it was run. We compute the slope in two ways, and they disagreed by more than our error bars allowed: the bars were too small. We enlarged them to include every reasonable way of doing the extrapolation.

The numbers: 36.8 hours of agent time over 13 days, 1.42 billion tokens processed, and 79 prompts from us. The effort went where you might not expect. Reproducing all the known solutions took 4.4 hours; the first genuinely new object, the rotating Gauss–Bonnet solver, took 11.2; and getting Ψ to zero temperature, with the paper and the referee rounds, another 8.7.

So our lesson: writing the code was cheap; verifying it was the work. Ask for checks that can fail, and let one agent referee the other. Everything is in the repository.



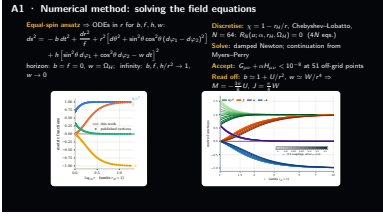
6. Thank you

0:05 [5:00]

Thank you for your attention, and thanks to Matías and Nahuel for the AI course.

Backup slides

Not part of the five minutes. Use them if there are questions; the notes below are written to be read, not recited.



A1. Numerical method: solving the field equations

backup

Why ordinary differential equations. A generic rotating black hole in five dimensions depends on the radius and on an angle. When the two angular momenta are equal, the extra symmetry removes the angular dependence: the metric on the slide is fixed up to four functions of the radius alone, b , f , h and w . The field equations of Einstein–Gauss–Bonnet gravity then become ordinary differential equations in r . Despite the curvature-squared term they stay second order, which is what makes Gauss–Bonnet special.

Boundary conditions. At the horizon, $r = r_H$, both b and f vanish and w equals the angular velocity of the horizon, Ω_H . Far away the metric must become flat space: b , f and h/r^2 go to one and w goes to zero. Each solution is labelled by three numbers: r_H , Ω_H and α . We set $r_H = 1$; every other size follows by rescaling.

Discretisation. The radius runs to infinity, so we use $\chi = 1 - r_H/r$, which maps the horizon to $\chi = 0$ and infinity to $\chi = 1$. Each function is represented by its values at $N = 64$ Chebyshev–Lobatto points, a spectral method: for smooth functions the error falls faster than any power of N . The equations become $4N$ algebraic equations $R_N = 0$ for the $4N$ unknown values.

Solving. Newton’s method, with damping for robustness. We start from the exact Einstein solution (Myers–Perry, $\alpha = 0$) and move step by step in α and in Ω_H , using the previous solution plus its derivative as the next initial guess.

Accepting a solution. A separate program, which does not use the discretised equations, computes the full curvature of the metric and checks the field equations at 51 points between the grid points. The residual must be below 10^{-8} relative to the size of the terms.

Reading off the physics. Far away, $b \approx 1 + U/r^2$ and $w \approx W/r^4$. The coefficients give the mass and each angular momentum, $M = -3\pi U/8$ and $J = \pi W/4$. The temperature comes from the slope of b and f at the horizon, and the entropy from the Wald formula, which for Gauss–Bonnet has an extra term proportional to α times the curvature of the horizon.

Right plot. The four functions at the 13 couplings, all at the same spin. The change with the coupling is not small: at the largest coupling the horizon is squashed in the opposite sense from the Einstein solution.

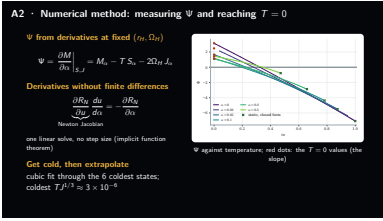
Left plot: step 0, reproducing the published family. Before measuring anything new, the solver had to reproduce what was known: the rotating Gauss–Bonnet solutions of Brihaye, Kleihaus, Kunz and Radu (2010), their Figure 1. The lines are ours, the circles are theirs. The curves were not digitised from the image: the arXiv source contains the figure as vector PostScript, so the agent read the exact vertices and calibrated the axes from the file’s own

tick marks. (Their coupling is $\alpha_{\text{ref}} = 4\alpha$ in our normalisation, fixed by matching the entropy formulas.)

2×10^{-4} *at the horizon*. The largest difference there is 2.07×10^{-4} , which is 0.4 of the smallest step their plot can resolve (5.2×10^{-4} in data units). The horizon values fix the parameters and the boundary conditions, so this says we solved the same problem.

About 10^{-2} in the interior. Away from the horizon the curves differ by up to 1.4×10^{-2} . The control is the curve at almost zero coupling in the same figure, where the exact solution is known: their plotted curve is 9.5×10^{-3} away from the exact formula. So their plot cannot test ours more sharply than about one percent; the origin of their offset is unknown.

One open number. An ergosurface radius printed in the paper’s text does not match: we get 1.1021, they quote 1.104, about four times their rounding error. The two other values of that series agree. We report it as an open discrepancy, not as an error of theirs. It does not enter Ψ or the extrapolation.



A2. Numerical method: measuring Ψ and reaching $T = 0$

backup

What Ψ is. Ψ is not a field; it is one number per solution. It is the change in mass when the coupling changes while entropy and angular momentum are held fixed, just as temperature is the change in mass per unit of entropy. The code, however, holds r_H and Ω_H fixed, and changing α that way also changes the entropy and the angular momentum. The first law tells us how to correct for that: subtract T times the change of entropy and $2\Omega_H$ times the change of angular momentum. What remains is the formula on the slide. The factor 2 counts the two equal angular momenta.

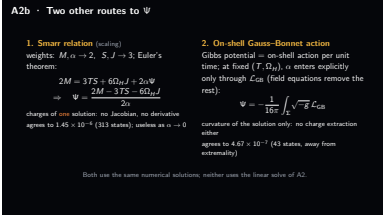
A closed-form example shows why the subtraction matters. For the static black hole, the mass at fixed radius grows with α (derivative $3\pi/4$), but at $\alpha \rightarrow 0$ the potential is $\Psi = -9\pi/4$: negative. Holding the radius fixed and holding the entropy fixed are different comparisons.

How we get the derivatives. The obvious approach is to solve at α and at α plus a small step, and subtract. That forces a choice of step: too large and the difference is inaccurate, too small and rounding errors dominate, and it gets worse near zero temperature. Instead we differentiate the discretised equations themselves. Since $R_N = 0$ holds along the whole family of solutions, its total derivative in α vanishes, which gives the linear system on the slide. Its matrix is the Jacobian that Newton’s method already built, so one extra linear solve gives $du/d\alpha$ exactly for the discrete problem, with no step size. This is the implicit function theorem. We checked it against ordinary finite differences: the difference shrinks by a factor of about four each time the step is halved, as it should.

One trap: the Wald entropy depends on α explicitly, not only through the metric. Forgetting that term gives a smooth but wrong answer, so it is coded and tested separately.

Reaching zero temperature. The code cannot be run exactly at $T = 0$, where the horizon

becomes degenerate. We push to very cold solutions, down to $TJ^{1/3} \approx 3 \times 10^{-6}$, fit Ψ against T with a cubic through the six coldest states, and read off the value at $T = 0$ (the red dots). Changing the fit (number of points, degree, square-root or logarithmic terms, T or $TJ^{1/3}$ as variable) gives the error bars of the result slide. That extrapolation is the largest uncertainty in the whole calculation.



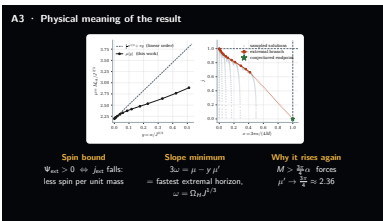
A2b. Two other routes to Ψ

backup

These are the two checks on the validation slide that do not use the linear solve of A2. Both start from the same numerical solutions.

Smarr. M and α scale like length squared, S and J like length cubed. Euler's theorem for a function of that kind gives $2M = 3TS + 6\Omega_H J + 2\alpha\Psi$, so Ψ follows from the charges of a single solution. The division by α makes it useless near $\alpha = 0$. Over 313 states it agrees with the main method to 1.45×10^{-6} .

On-shell action. At fixed temperature and angular velocity the Gibbs potential is the on-shell action per unit time. The only explicit α in the action is the Gauss–Bonnet term, and the implicit change of the fields drops out by the field equations, so Ψ is minus the Gauss–Bonnet density integrated outside the horizon, over 16π . It needs no charges at all. On 43 states, none of them near extremality, it agrees to 4.67×10^{-7} .



A3. Physical meaning of the result

backup

The sign: the spin bound tightens. A positive slope means the minimum mass at fixed spin grows with the coupling. Equivalently, the maximum spin a black hole of given mass can carry, j_{ext} , decreases (right panel, red curve, which starts at $j = 1$, the Einstein bound). Gauss–Bonnet black holes can spin less than Einstein ones.

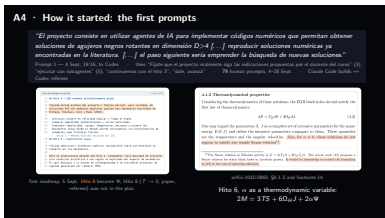
A reading through the Smarr relation. At zero temperature the Smarr relation becomes $\alpha\Psi_{\text{ext}} = M_{\text{ext}} - 3\Omega_H J$. In Einstein gravity the right-hand side is exactly zero. A positive slope says that the Gauss–Bonnet term opens a gap between the mass and the rotational energy term, and our measurement says this gap keeps opening, at a rate that first falls and then recovers.

Where the slope is smallest. Writing the same relation in scale-free variables gives $3\omega = \mu - y\mu'$, with $\omega = \Omega_H J^{1/3}$ the dimensionless horizon angular velocity. Differentiating, the minimum of the slope μ' is exactly the maximum of ω : the turning point is the extremal black hole whose horizon rotates fastest. The data confirm this independently.

Why the slope must rise again. Static Gauss–Bonnet black holes, and also the rotating ones, satisfy $M > 3\pi\alpha/4$. If the slope stayed near 1, the extremal mass would eventually fall below that bound. So the slope has to climb back to at least $3\pi/4 \approx 2.36$. The rise we see from 1.07 to 1.44 is therefore required, not accidental. If, as conjectured by Kleihaus, Kunz and Radu, the extremal branch ends at the singular static configuration (the green star on the right, $x = 1$), the slope would approach 2.36. We only reach $x \approx 0.42$, so this is consistent with the data but not shown by them.

The sign of Ψ in geometric terms. Ψ equals minus the Gauss–Bonnet curvature density integrated over the space outside the horizon, divided by 16π . Positive Ψ means that integral is negative. It is negative for slowly rotating black holes and positive near extremality.

Limits of the statement. Classical Einstein–Gauss–Bonnet gravity, equal spins, one branch of solutions (the one connected to Myers–Perry), a finite range of couplings. In an effective-field-theory view, other higher-curvature terms enter at the same order as α^2 , so the non-monotonic slope is a property of this theory, not a universal prediction.



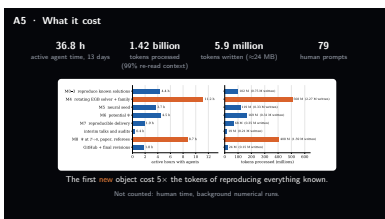
A4. How it started: the first prompts

backup

The project began on 4 September as the final project of this course. At the top is the very first prompt, to Codex: write numerical codes for rotating black holes in more than four dimensions, reproduce the literature, then look for new solutions. Later prompts were mostly short: “continue with milestone 3”, “go ahead”.

On the left, the roadmap the agent wrote on day two. Milestone 4, reproducing Brihaye et al., was the “minimum success”. Milestone 6, a “new exploration” to be chosen later, became the potential Ψ . Milestone 8, zero temperature and the paper, was not planned at all.

On the right, the reproduced paper. For $\alpha \neq 0$ it finds no simple Smarr relation, and footnote 14 suggests extending the static one of Kastor, Ray and Traschen. Treating α as a thermodynamic variable, scaling gives it: $2M = 3TS + 6\Omega_H J + 2\alpha\Psi$, which our solutions satisfy to about 10^{-7} .



A5. What it cost

backup

About 37 hours of active agent time over 13 days, and 79 human prompts. The agents processed 1.4 billion tokens, but 99 percent of that is context being re-read at every tool call; they *wrote* about six million tokens, some 24 megabytes of text. The expensive blocks, in orange, are building the rotating solver and the final measurement with the paper and the referee rounds. Reproducing all known solutions took about four hours; the first new object took five times as

many tokens, and five solver strategies were discarded before one worked. Not counted: our own time and the numerical runs in the background.

If there are questions

Why is J not like a charge, for the weak gravity conjecture?

Angular momentum is not a gauge charge, so the WGC spectrum arguments do not carry over. We use the Goon–Penco relation only as an identity of the first law, and draw no WGC conclusion.

Why not solve directly at $T = 0$?

Kleihaus, Kunz and Radu (2023) did build extremal solutions directly, in a different gauge. Our solver works at a fixed horizon radius and angular velocity, so we approach $T = 0$. The spread over the fitting choices is reported as a sensitivity, not an error bar. A direct construction is the natural extension: a solver with a degenerate horizon, seeded by our coldest states, would remove the extrapolation, though it would share our field equations and mass extraction.

Can’t their extremal solutions remove the extrapolation caveat?

In principle yes, since they involve no extrapolation in temperature. But the paper prints no numbers: the extremal branch appears only as curves in their Fig. 1, normalised by the mass, with no stated accuracy. Read off a plot, the masses are good to about one percent; our mass spread is 10^{-3} to 10^{-6} , and a slope from differences of such masses would be uncertain by order one. Their raw data would test our zero-temperature masses directly. From the figure, only a rough check of the sign and the $2.3\times$ overshoot is possible.

What exactly is new?

Ψ on rotating solutions at finite coupling; the slope at 13 couplings, with its minimum and its rise; the extremal mass curve and how far it departs from first order; and the reading of Ψ as an on-shell Gauss–Bonnet action.

Which parts did the humans do?

We chose the problem and the reference papers, steered each milestone, asked for critical reviews, made the agents referee each other, and questioned the results. The agents wrote the code, ran it and drafted the text.

On crediting the AI

AI tools cannot be authors, because they cannot take responsibility for the work. That is the position of COPE and of Springer Nature, which publishes JHEP; arXiv likewise holds the human authors responsible for everything submitted. Springer Nature asks for a disclosure in the methods section or the acknowledgements, giving each tool’s name, version, maker and purpose. The title slide carries the short form, “with Claude Code and Codex”. For the paper, a sentence such as: “Code, numerical computations and a first draft were produced with Claude Code (Anthropic) and Codex (OpenAI) under the authors’ direction; the authors checked the results and take full responsibility for the content.”